Single-molecule multimodal timing of in vivo mRNA synthesis

JC notes

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1 Abstract

- Poly(A) tail measurement to distinguish transcribing vs. poly(A) transcripts.
- Splicing is much more delayed than previously tought, for at least 10 kb.
- Splicing mostly apparent only in poly(A) transcripts $\rightarrow$ they think it's delayed mostly after 3' end formation.
- Lots of m^6A on unspliced RNA $\rightarrow$ deposition of m^6A happens before splicing.
- m^6A supressed around exon boundaries $\rightarrow$ m^6A topology should be established before EJC deposition.
- 3' end cleavage occurs multiple kbs behind transcription and is coordinated with transcription termination. The 3' end formation is recursive in many genes.
- Overall, they think that cleavage, termination and m^6A deposition occur earlier than most of the splicing.

2 Intro

- Spliceosome assembly is known to be rapid.
- Splicing kinetics: estimates vary across methods.
- Previously, it was tought that EJC prevents m^6A deposition at exon boundaries. Here, they argue that m^6A is placed before splicing (and thus, before EJC placement).
  - m^6A methylation machinery is known to interact with Pol II $\rightarrow$ permitting co-transcriptional methylation.

Poly(A) notes

- Poly(A) site: cannonical AAUAAA motif.
- When Pol II reaches it, it keeps transcribing.
- Protein complex cut RNA there; poly(A) polymerase (PAP) adds   200 As.
- Transcription termination:
  - Torpedo model: Exonuclease XRN2 start chewing the 5' end of RNA still transcribed by Pol II → when it catches Pol II, it knocks it of DNA.
  - Allosteric model: Pol II changes conformation at poly(A) site, becomes less stable and eventually falls of DNA.

3 Results

3.1 Development of a multimodal direct pre-mRNA sequencing technique

- Need to distinguish whether pre-mRNA has a poly(A) tail:
  - pre-mRNA^{e}: undergoing transcription.
  - pre-mRNA^{a}: polyadenylated pre-mRNA, undergoing post-transcriptional maturation.
- They pulled Pol II with associated pre-mRNA via immunoprecipitation. The pre-mRNA^{a} was still somehow pulled (even thought it was already cleaved - otherwise it would not have the poly(A) tail); probably was somehow attached via the cleavage complex, or some chromatin components.
- They got about 90% of pre-mRNA^{e} and 10% pre-mRNA^{a}.
- 95% of pre-mRNA^{e} have unspliced introns, while this is only 58% for pre-mRNA^{a}.
- Nanopore sequencing requires poly(A) tail → added artificially during library preparation. This *in vitro* poly(A) tail is shorter (20-80nt), so they get bimodal distribution of tail lengths, and classify by cut-off.
  - Also, the reads classified as pre-mRNA^{a} have the start of poly(A) highly enriched by known poly(A) sites.